

GROUP 14

EDGE-BASED ANOMALY DETECTION FOR FLEET BASED CONDITION MONITORING OF MACHINES

PROJECT FOCUS:

Early fault detection in machine fleets before failure occurs

KEY INNOVATION:

- Fleet-based approach leveraging similar machines
- Unsupervised anomaly detection without historical labeled data
- Real-time online monitoring with visualization

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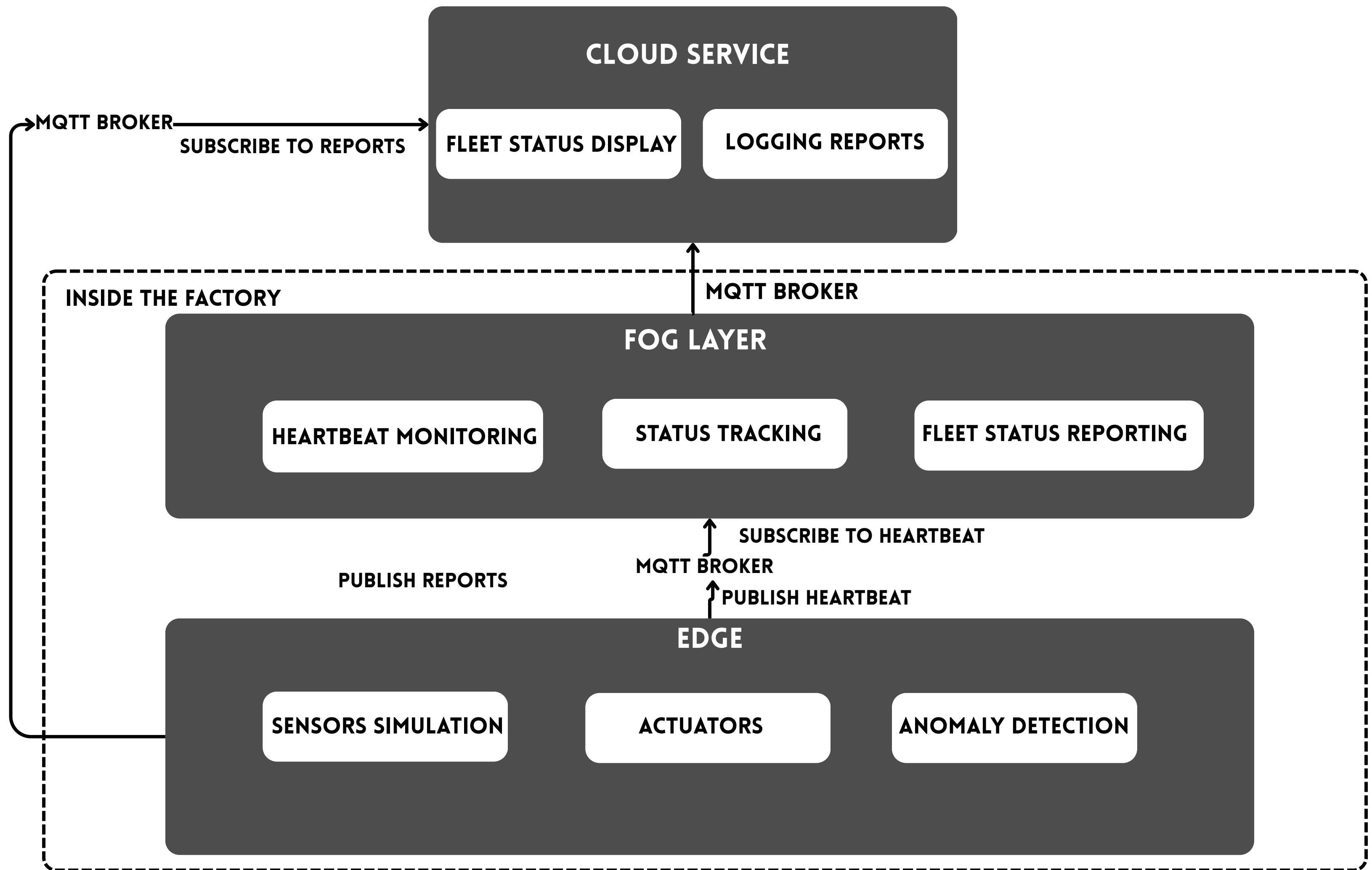


PROBLEM STATEMENT

Machine failures decrease up-time and can lead to extra repair costs or even to human casualties. Recent condition monitoring techniques use artificial intelligence in an effort to avoid time-consuming manual analysis and handcrafted feature extraction. Many of these only analyze a single machine and require a large historical data set. In practice, this can be difficult and expensive to collect.

Processing all the data at the cloud will require the data to be transmitted to cloud and then process it, thereby increasing the total latency to identify the fault.

Hence, we designed a edge-fog-cloud architecture that brings the fault-detection to the edge node, and a unsupervised model that removes the need for large historical dataset.



Fleet Anomaly Dashboard (Edge→Fog→Cloud)

Fleet Snapshot

Timestamp: 2025-10-23T03:10:34.349330Z | Edge nodes: 3

Node Status:

	edge_id	status	n_machines
0	edge2	alive	100
1	edge3	alive	100
2	edge1	alive	100

Anomalous Machines per Edge Node:

	edge_id	anomalous_machines
edge2	edge2	8 32 35 48 94 97
edge3	edge3	85
edge1	edge1	3 7 13 14 20 30 41 48 56 74 95

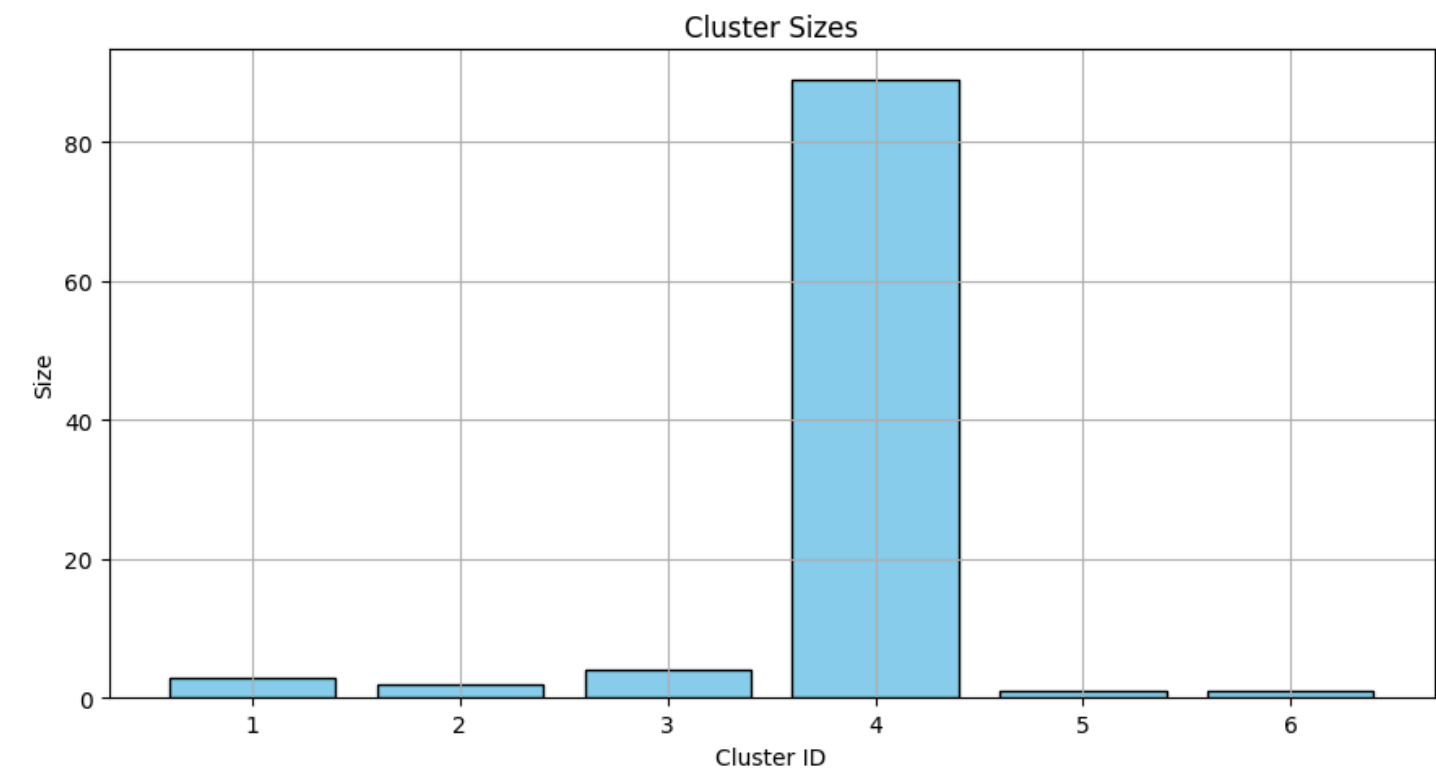
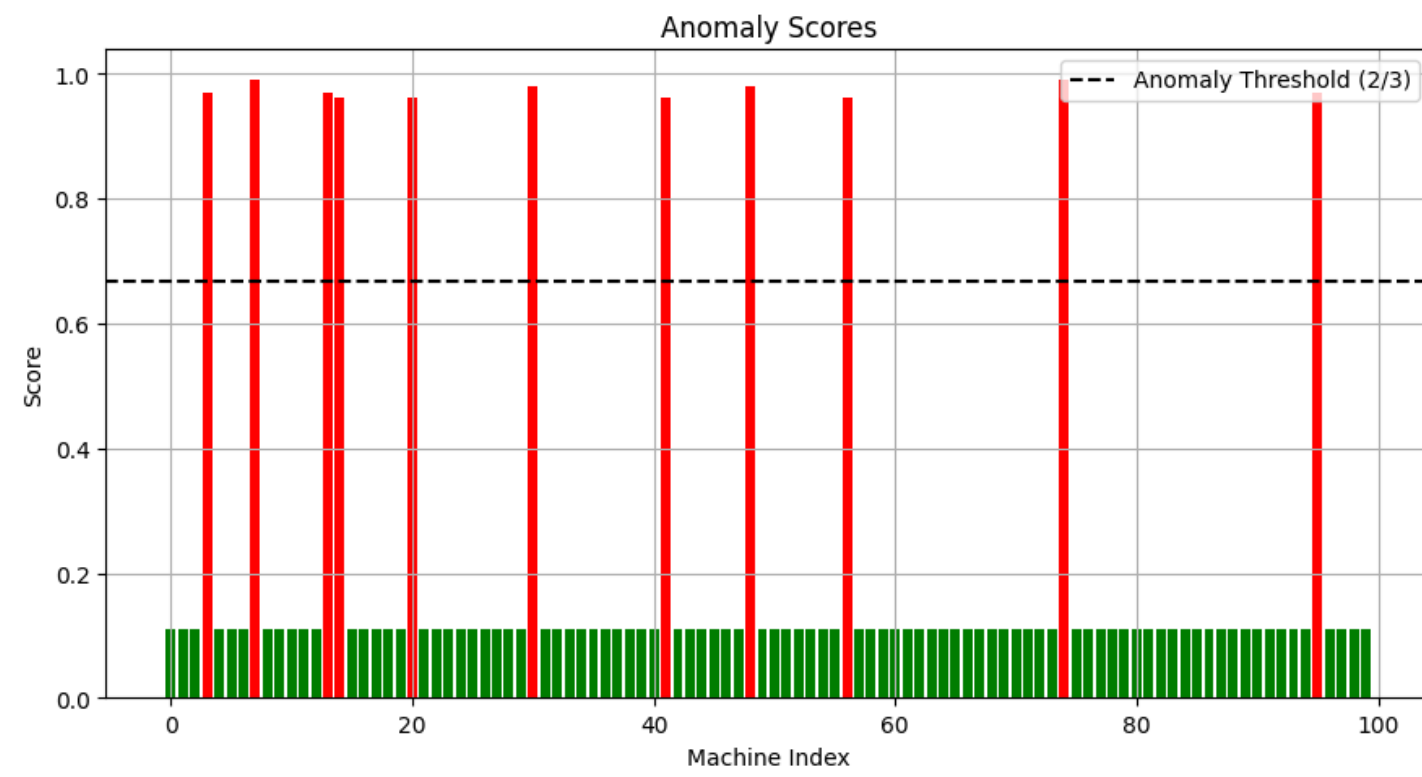
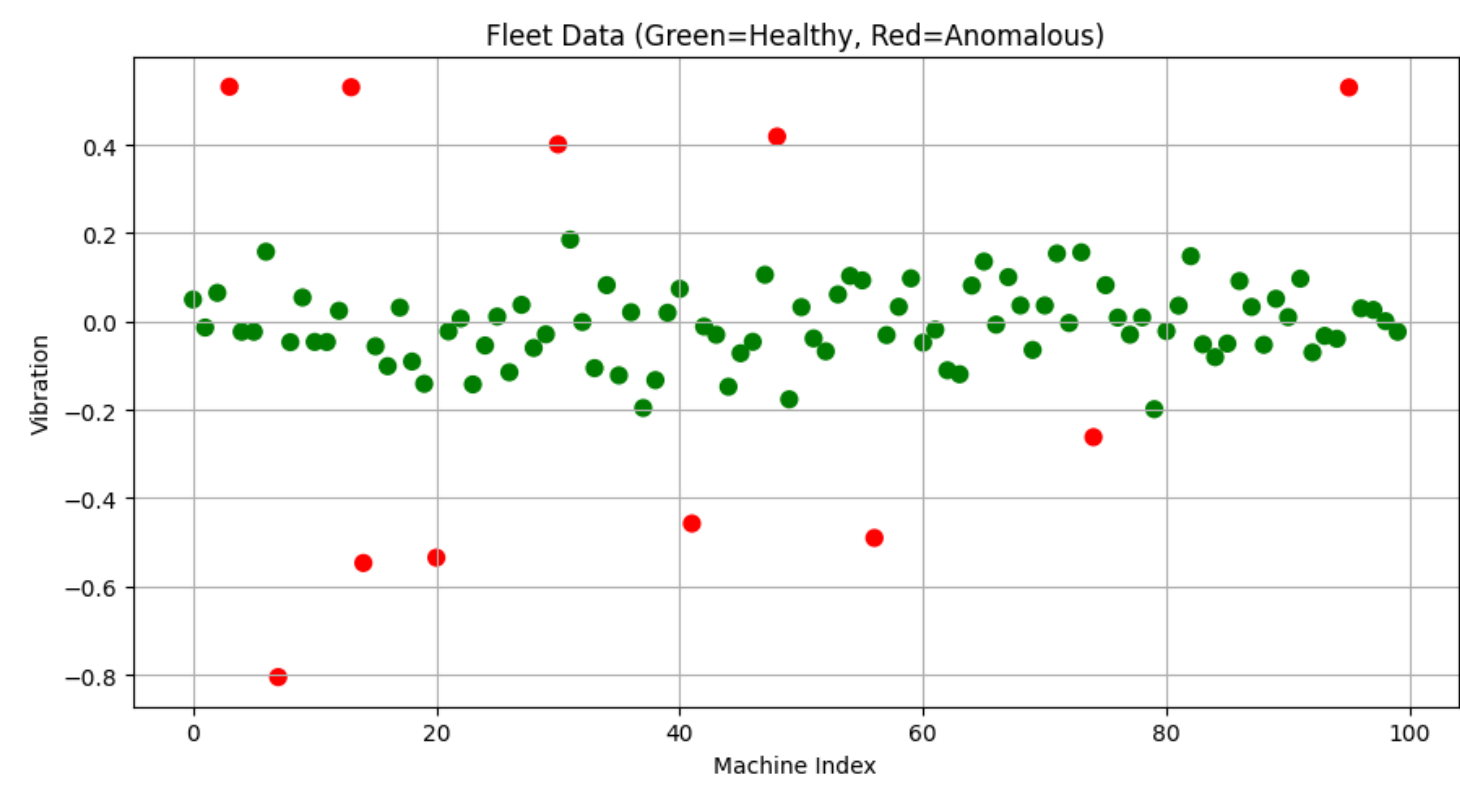
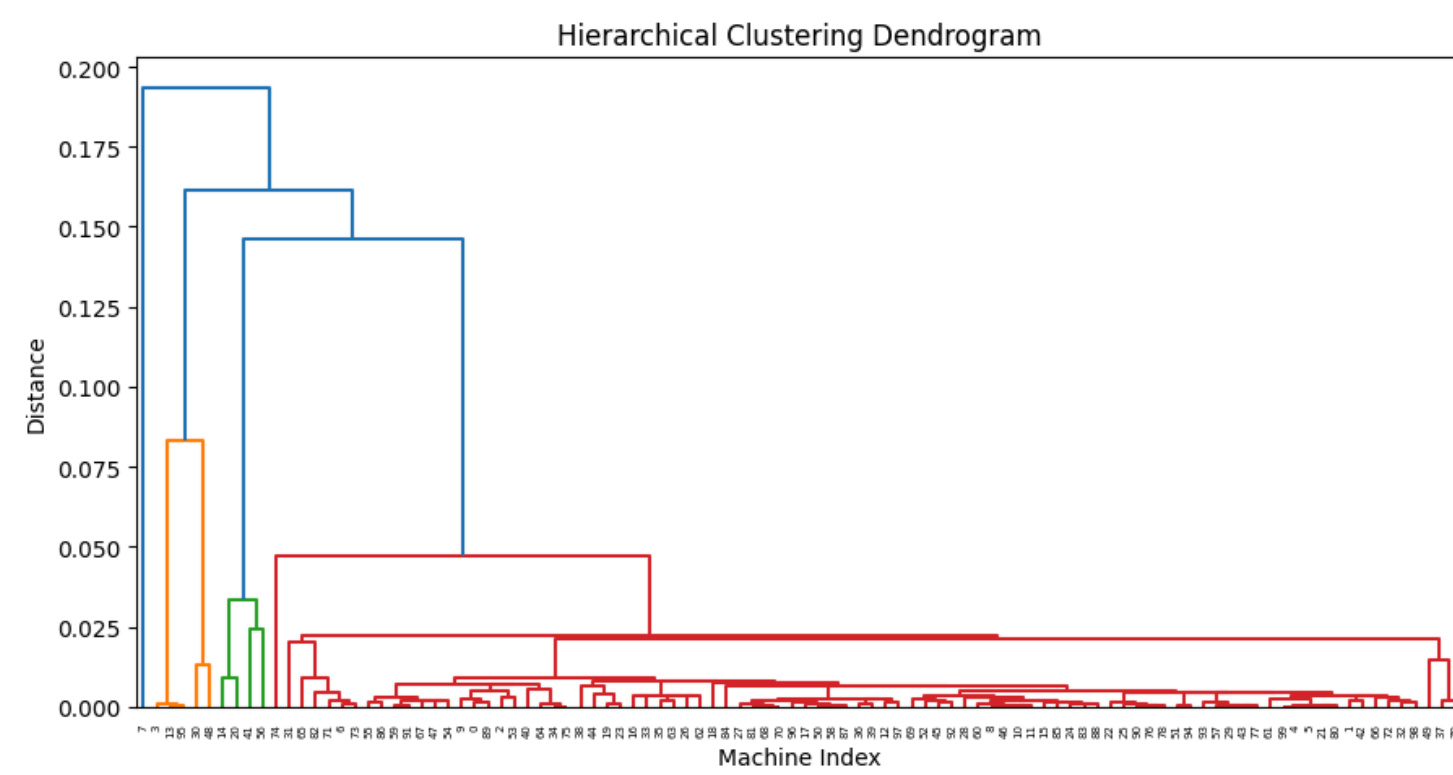
Live Report Stream

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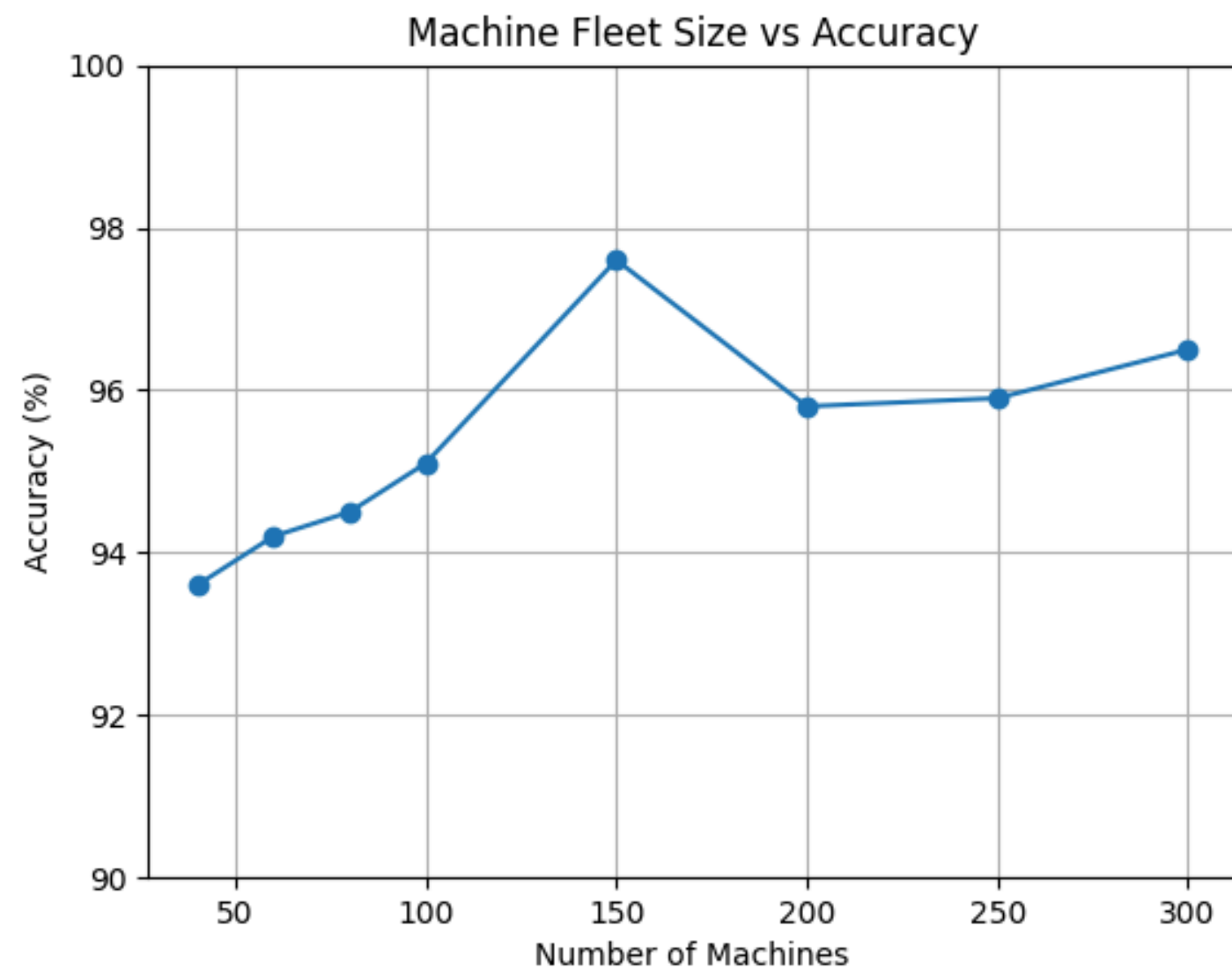
	edge_id	ts	metrics			
0	edge3	2025-10-23T03:10:31.129662Z	-0.0091949919865191	-0.1463350652811679	0.1081791679198325	-0.0239325
1	edge2	2025-10-23T03:10:31.129189Z	-0.0348894450989425	0.0983703433351799	0.058092283071714	0.007028444
2	edge1	2025-10-23T03:10:31.128984Z	0.0496714153011232	-0.0138264301171184	0.0647688538100692	0.53152257
3	edge3	2025-10-23T03:10:11.101582Z	-0.0091949919865191	-0.1463350652811679	0.1081791679198325	-0.0239325
4	edge2	2025-10-23T03:10:11.100801Z	-0.0348894450989425	0.0983703433351799	0.058092283071714	0.007028444
5	edge1	2025-10-23T03:10:11.100264Z	0.0496714153011232	-0.0138264301171184	0.0647688538100692	0.53152257
6	edge3	2025-10-23T03:09:56.078886Z	-0.0091949919865191	-0.1463350652811679	0.1081791679198325	-0.0239325
7	edge2	2025-10-23T03:09:56.078476Z	-0.0348894450989425	0.0983703433351799	0.058092283071714	0.007028444
8	edge1	2025-10-23T03:09:56.078252Z	0.0496714153011232	-0.0138264301171184	0.0647688538100692	0.53152257
9	edge3	2025-10-23T03:09:41.062244Z	-0.0091949919865191	-0.1463350652811679	0.1081791679198325	-0.0239325
10	edge2	2025-10-23T03:09:41.060800Z	-0.0348894450989425	0.0983703433351799	0.058092283071714	0.007028444
11	edge1	2025-10-23T03:09:41.059517Z	0.0496714153011232	-0.0138264301171184	0.0647688538100692	0.53152257
12	edge2	2025-10-23T03:08:55.995461Z	-0.0348894450989425	0.0983703433351799	0.058092283071714	0.007028444



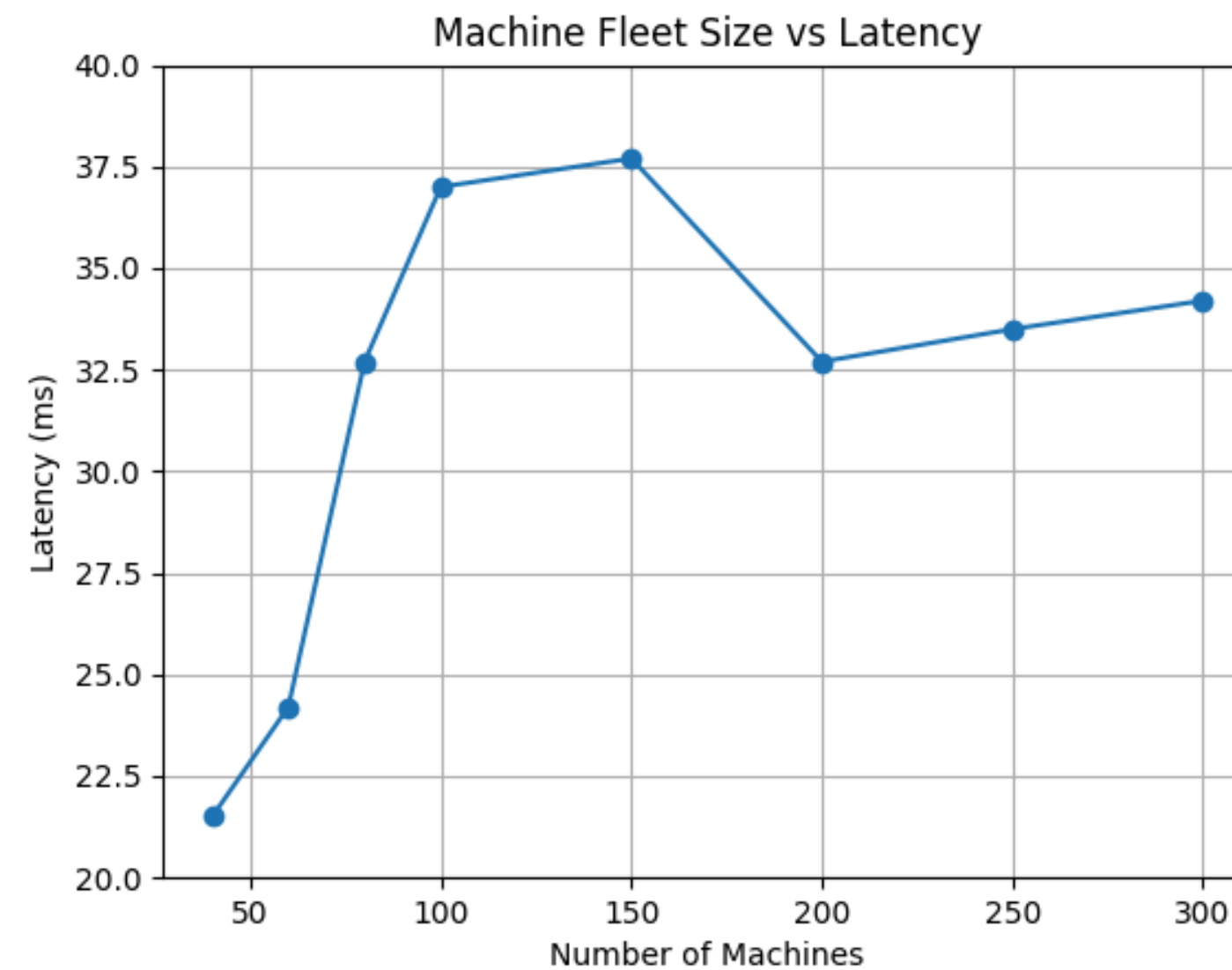
$$\text{Anomaly Score}_i = 1 - \frac{\text{Size of Cluster}_i}{\text{Total Number of Machines}}$$

$$\text{Machine}_i = \begin{cases} \text{Normal,} & \text{if Anomaly Score}_i < \frac{2}{3}, \\ \text{Anomalous,} & \text{otherwise.} \end{cases}$$

ACCURACY



END-END LATENCY



PROCESSING POWER

Host device specifications for docker containers:
Intel i7 processor with 16GB RAM.

Edge:

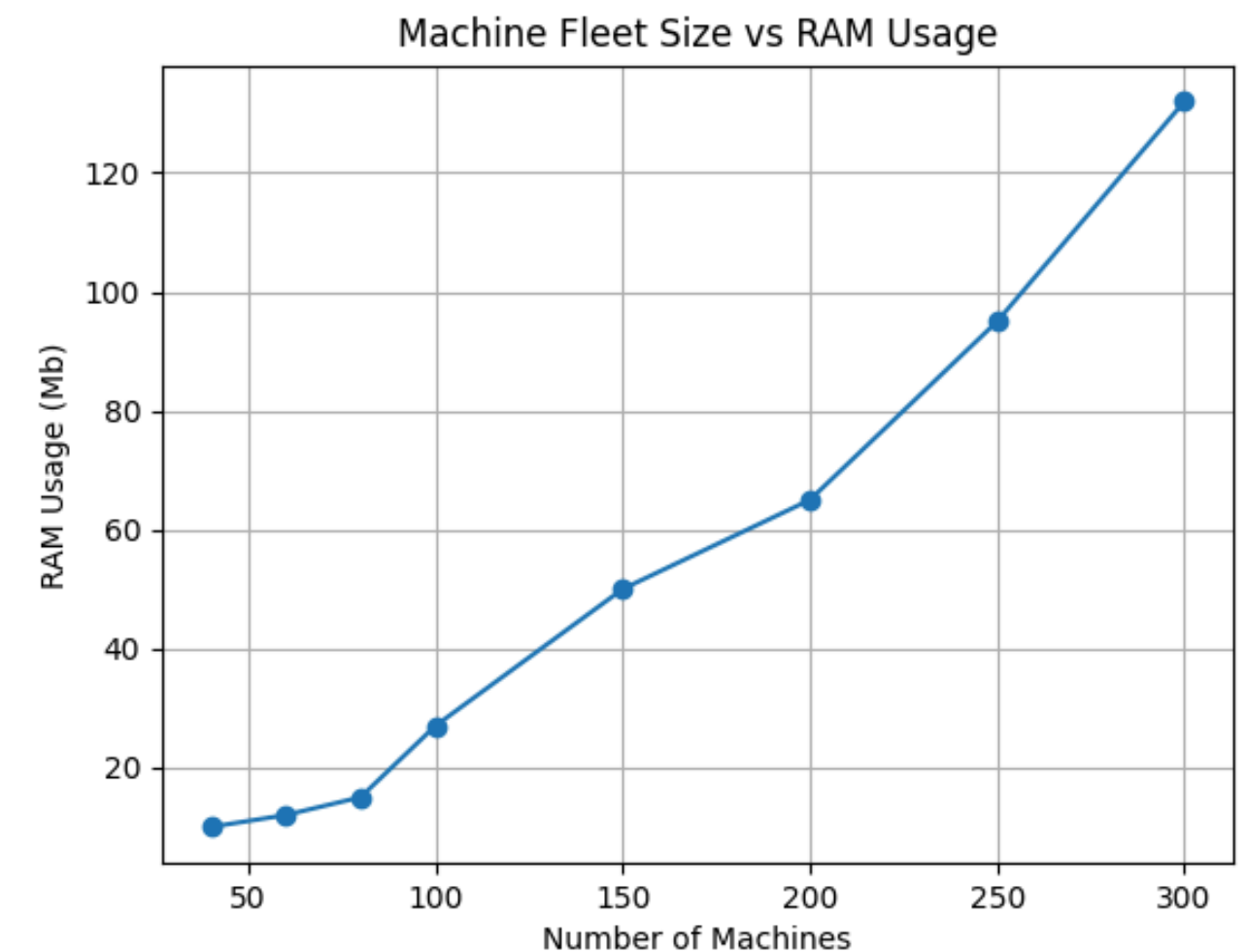
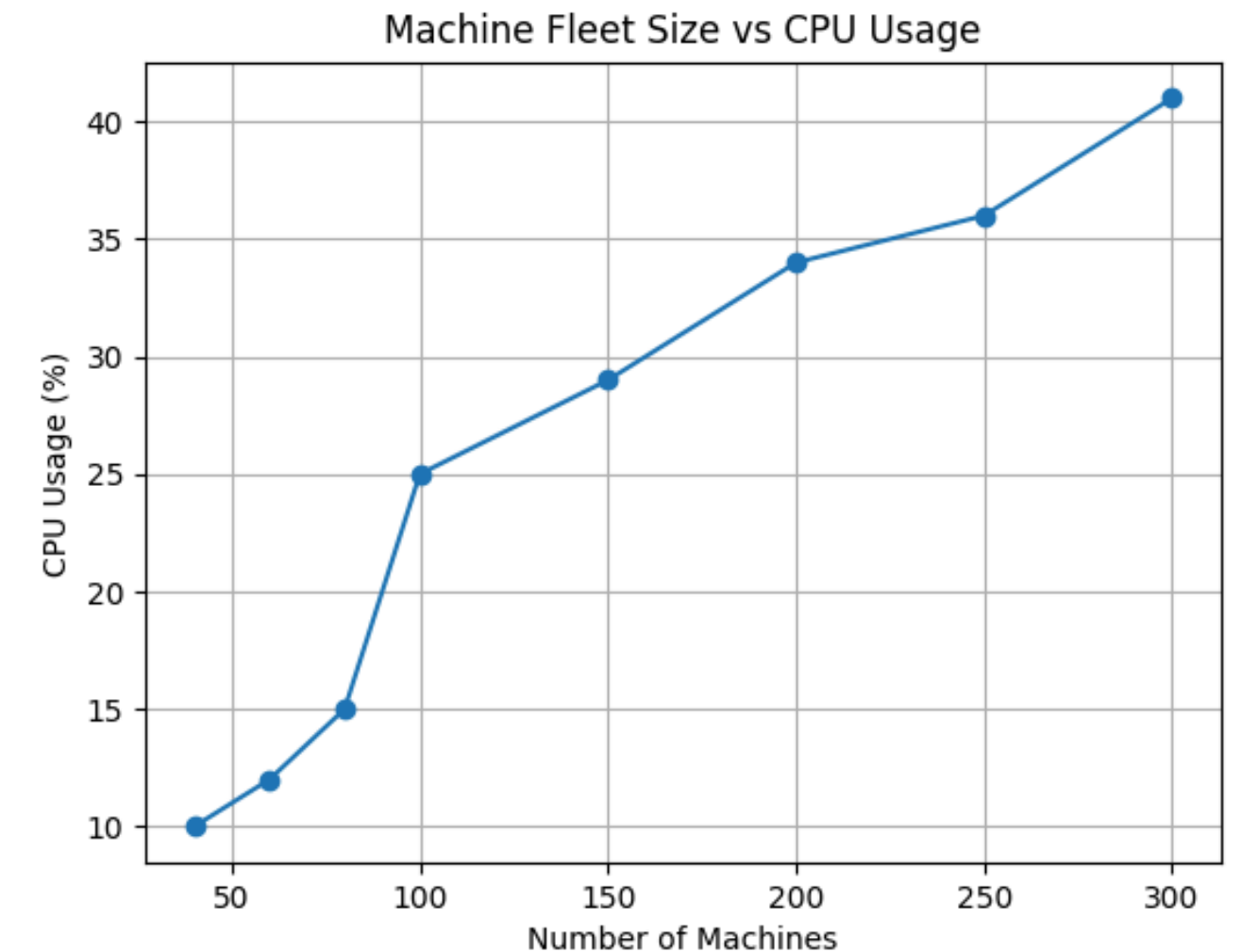
- 60-70MB is consumed by docker.
- 0.5 CPU time of the host machine.

Fog:

- 512MB RAM allocated.
- 1 CPU time of the host machine.

Cloud:

- No limitations on RAM and CPU is imposed.



ARCHITECTURE & DATA FLOW

- **Edge-Centric Design:** All intelligence and data processing (e.g., analysis, filtering) reside directly on the edge nodes.
- **Cloud Dashboard:** The cloud service (using Streamlit) acts as a visualization layer, responsible only for displaying the final results.
- **Targeted Data Flow:** Edge nodes publish processed insights or key readings (not raw data) via MQTT to the cloud for display.

FUZZY INFERENCE SYSTEM (FIS) LOGIC

- **Inputs:** The system uses two key metrics calculated from the data :
 - **Max Z-Score (Severity of spikes) .**
 - **Mean Z-Score (Overall stability) .**
- **Rules:** It applies intuitive, human-readable rules like:
 - IF MaxZScore is High OR MeanZScore is Erratic THEN Anomaly is Critical .
- **Output:** A nuanced AnomalyScore between 0 (Healthy) and 1 (Critical).

$$\mu(x) = \max \left(0, \min \left(\frac{x-a}{b-a}, \frac{c-x}{c-b} \right) \right)$$

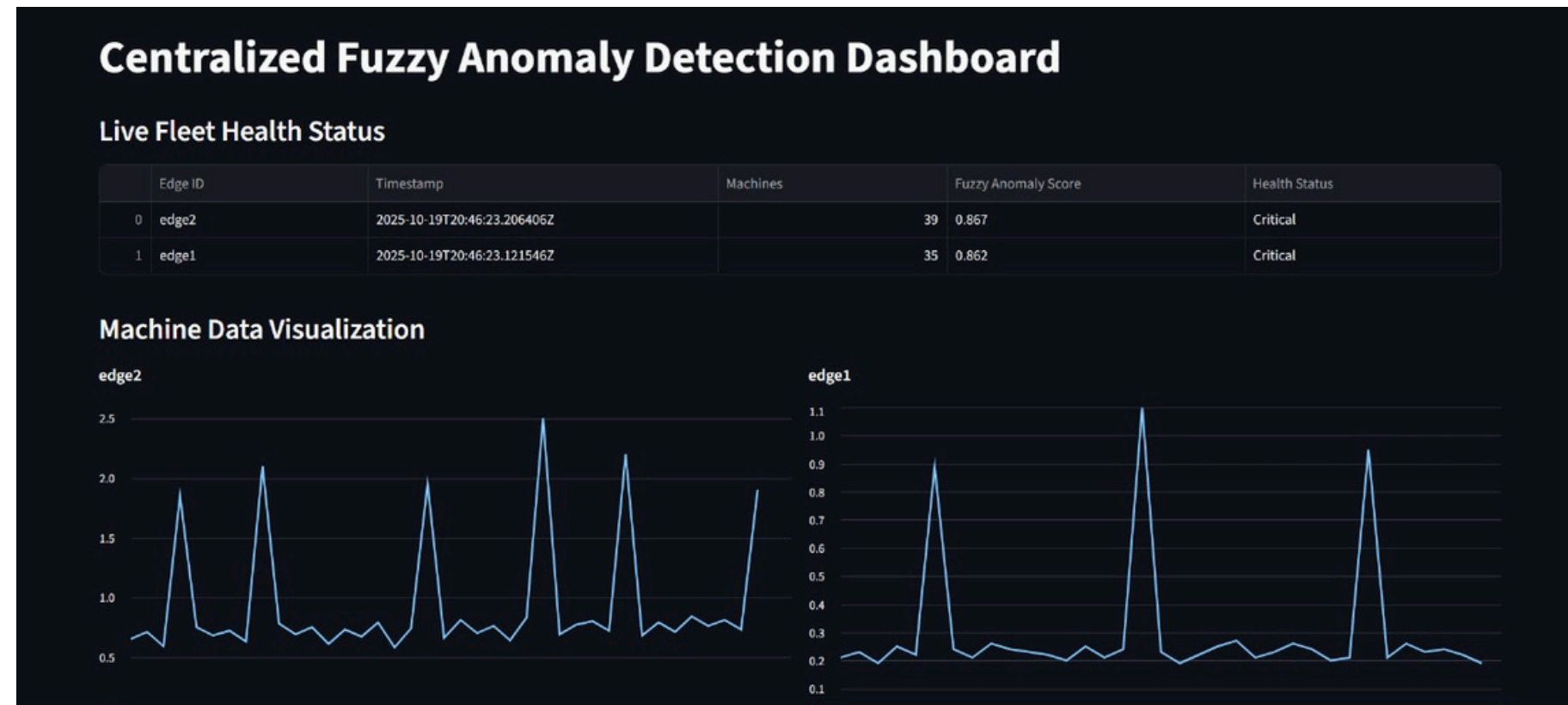
x: Input value

a: The left starting point (membership = 0)

b: The center peak (membership = 1)

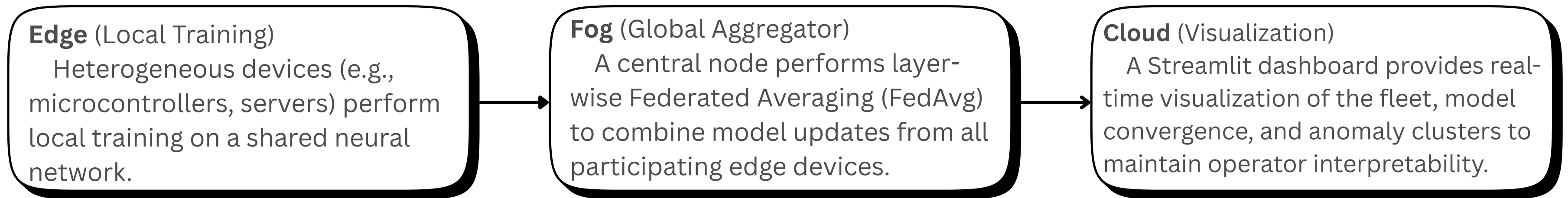
c: The right ending point (membership = 0)

FUZZY PROTOTYPE: RESULTS & LIMITATIONS



- **Centralization Bottlenecks:** Streaming all raw data consumed significant bandwidth and created a single point of failure .
- **Static Rules:** The fuzzy rules were not adaptive and required manual re-tuning for new machine behaviors or fault types .
- **Federation Incompatibility:** The design was inherently incompatible with the project's goal of creating a privacy-preserving, federated system.

ME-FLEET: ARCHITECTURE & METHODOLOGY



- **Key Principles:**

1. **Multi-Exit Models:** Allows resource-constrained devices to train and contribute updates for shallower parts of the model, while powerful devices train deeper layers.
2. **Layer-wise Aggregation:** The fog robustly merges partial updates by averaging weights for each layer independently, using only the contributions available for that layer.
3. **Privacy-Preserving:** Only model weights are shared, never the raw sensor data.

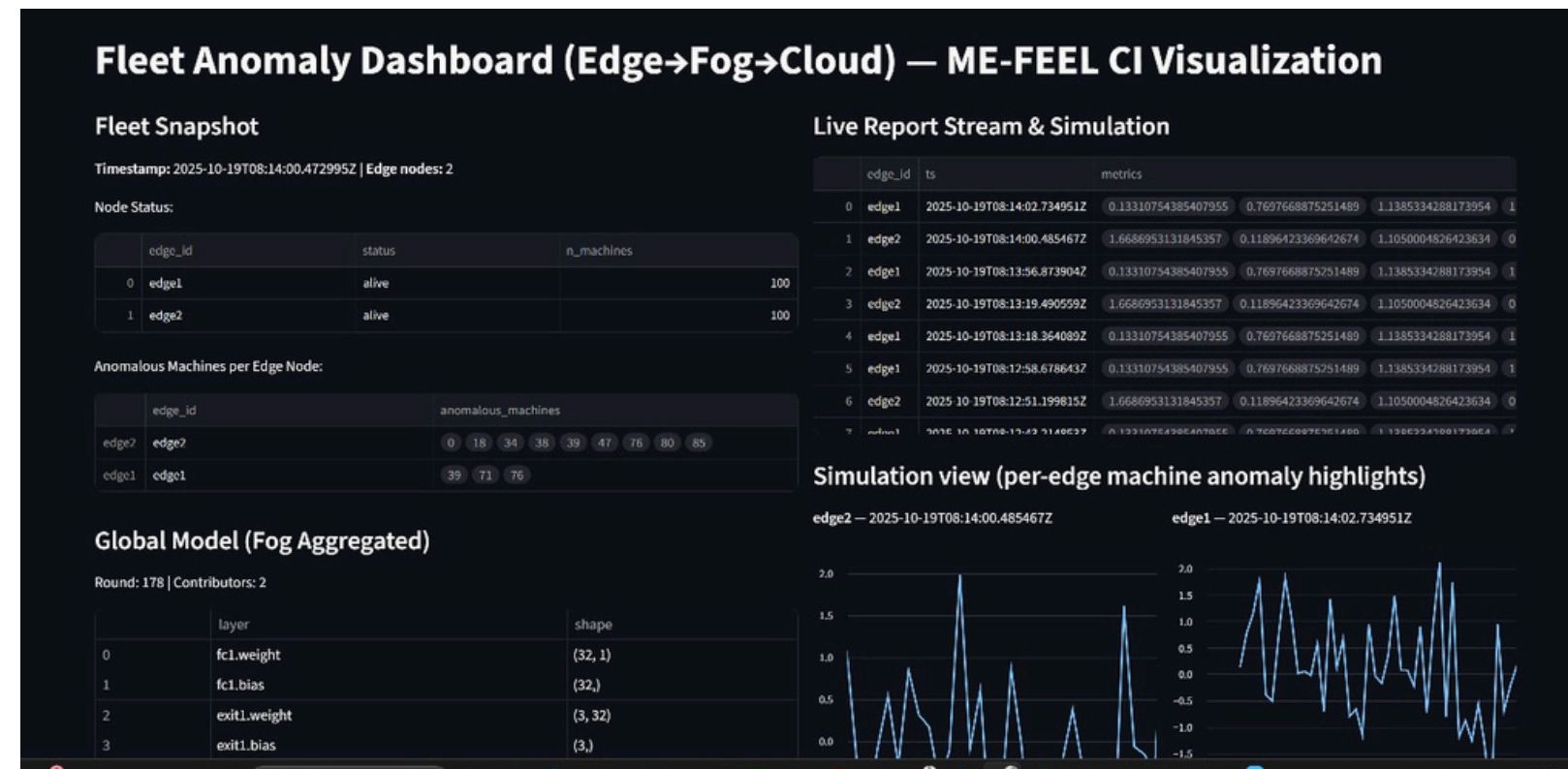
FEDERATED AGGREGATION STRATEGY

- In each aggregation round, the fog node computes a layer-wise weighted average of model updates from all clients.
- The update rule is given by:

$$w_l^{(t+1)} = \frac{\sum_{c \in C_l} |D_c| \cdot w_{c,l}^{(t)}}{\sum_{c \in C_l} |D_c|}$$

- WHERE:
 - $w_l^{(t+1)}$: Global weight for layer L at round (t+1)
 - C_l : Clients contributing updates for layer L
 - D_c : Local data size of client C
 - $w_{c,l}^{(t)}$: Local weight of client C for layer L

ME-FLEET: RESULTS & ADVANTAGES



- **Improved Accuracy:** Achieved ~90% detection accuracy, a +50% improvement over the fuzzy model's ~60% .
- **Reduced Communication:** Sharing weights instead of raw data led to a ~40% reduction in communication overhead .
- **Adaptive Learning:** The CI model can adapt to new fault modes, unlike the static fuzzy system .
- **Collaborative Intelligence:** Creates a shared, fleet-wide intelligence model without centralizing sensitive data.

ME-FLEET successfully balances adaptability and performance while retaining interpretability through its visualization dashboard.

INDIVIDUAL CONTRIBUTION – EDGE

Dwarakesh Venkatesh:

- Data generation
- Unsupervised Learning - Pairwise distance calculation
- Data visualization and model evaluation

Shyam Sundar Raju:

- Fog layer
- MQTT protocol

Sarigasini M:

- Unsupervised Learning - Agglomerative clustering
- Anomaly score calculation

Pooja Sree:

- Cloud layer
- Heartbeat mechanism

INDIVIDUAL CONTRIBUTION – CI

Aswath Siddharth R:

- Created Simulation in Docker for Multi Exit Federated Edge Learning for fleet based anomaly detection.

Darshan R:

- Created Simulation in Docker for Fuzzy based anomaly detection in fleet of machines